STAT3888: Post-Feedback Causal Inference

AUTHOR
Calvin Lai

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This document is mainly just to generate the new DAG based on our new list of potential counfounders, which are `AlcoholRisk`, `SmokingDuration`, `Sex`, `NonSchoolQuals`, `IncomeDecile`, `Age`, and `Waist`; the latter two being numerical while the others are categorical. As before, our main predictor is `ExerciseGuidelines` and our response is `ALT` or `GGT`.

▼ Code

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE,
                      fig.align='center', fig.dim=c(10, 8))

library(tidyverse)
library(here)
library(knitr)
library(dagitty)
library(ggdag)
```

Overall DAG Plot

Assuming all stated variables above are confounders, our DAG is as follows.

▼ Code

```
# DAG Plot
dag = dagitty("
dag {
  ExerciseGuidelines -> ALTGGT

  Age -> ExerciseGuidelines
  Age -> ALTGGT

  Sex -> ExerciseGuidelines
  Sex -> ALTGGT

  Waist -> ExerciseGuidelines
  Waist -> ALTGGT

  AlcoholRisk -> ExerciseGuidelines
```

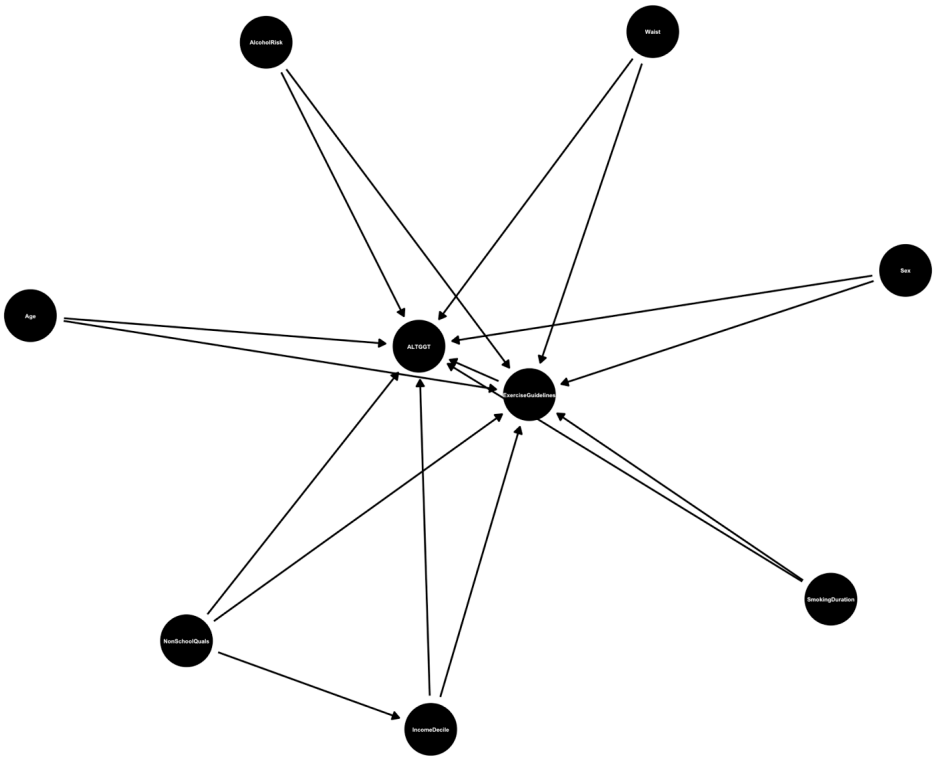
```
  AlcoholRisk -> ALTGGT

  SmokingDuration -> ExerciseGuidelines
  SmokingDuration -> ALTGGT

  NonSchoolQuals -> ExerciseGuidelines
  NonSchoolQuals -> ALTGGT
  NonSchoolQuals -> IncomeDecile

  IncomeDecile -> ExerciseGuidelines
  IncomeDecile -> ALTGGT
}
")

ggdag(dag, text_size = 1.35) + theme_dag()
```



▼ Code

```
adjustmentSets(
  dag,
  exposure = "ExerciseGuidelines",
  outcome = "ALTGGT"
)
```

```
{ Age, AlcoholRisk, IncomeDecile, NonSchoolQuals, Sex,
SmokingDuration,
  Waist }
```

Next, we should search for any literature that suggests that any of these variables are not actually confounders. Before, we have established that `Waist`, `Age` and `Sex` are confounders, so all that is left is to check whether `NonSchoolQuals`, `IncomeDecile`, `AlcoholRisk` and `SmokingDuration` are confounders, or some other type of variable (e.g., mediator, collider, or proxy); then, we may want to do some sensitivity analysis for those we deem other than confounder.

Causal Inference Analysis

I believe that the other four variables should remain as confounders, which I provide more details in the `CausalInferenceSources.pdf` document. For now, I suppose we keep the DAG as is, and I also save it for future use.

Hence, we can say the following for the report.

> **Important**
>
> Covariates were selected based on a causal DAG informed by prior literature. Age, sex, waist circumference, smoking history, alcohol-risk level and socioeconomic position were considered potential confounders of the physical activity–liver biomarker association. Educational qualification and income decile were used as available proxies for socioeconomic position
>
> Because the temporal relationship between physical activity and alcohol consumption is not fully established, alcohol-risk level was treated as a potential behavioural confounder; estimates should therefore be interpreted as associations with physical activity conditional on alcohol consumption.